TECHNICAL ASSESSMENT REPORT

# Egg Segmentation & Real-Time Video Tracking

A Comparative Study of Foundation Models vs. Classical Image Processing

**Task 4 Deliverable - Google Drive Video Tracking Links:**

Input Video: https://drive.google.com/file/d/1VhZXbqQUeHSfHX41rsnuAWxlQTsljW1C/view?usp=sharing

**Output Video: https://drive.google.com/file/d/12Lm6lshvtxvqnq_h5EeJbW2ggkfVuWfe/view?usp=sharing**

## 1. Project Overview & Approach

In this project, I developed an end-to-end computer vision solution to detect, segment, and track eggs across challenging real-world scenarios (varying illumination, occlusions, textured backgrounds, and high-speed motion). The work spans four core tasks:

* Task 1: Building a foundation-model pipeline (Grounding DINO + SAM) to generate reference pseudo-ground truth on 110 images.
* Task 2: Designing a purely classical image processing pipeline from scratch across 9 systematic trials, achieving 0.654 Eval IoU.
* Task 3: Evaluating classical predictions against SAM, conducting error analysis, and diagnosing physical failure causes.
* Task 4: Deploying the classical algorithm into an online multi-object tracker that tracks spinning eggs in video at 27.6 FPS.

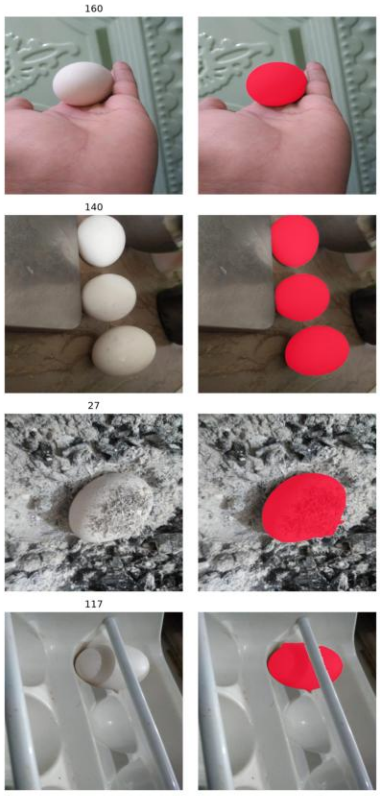
## 2. Task 1: Generating Ground Truth Using Pretrained Models

Why Foundation Models? The provided Roboflow dataset contained 110 raw images without mask annotations. To evaluate classical methods quantitatively in later tasks, I first established high-fidelity reference annotations using a two-stage zero-shot architecture:

1. Detector: Grounding DINO (IDEA-Research/grounding-dino-base) prompted with 'an egg.' at box threshold 0.30.
2. Segmenter: SAM (facebook/sam-vit-base) using detected boxes as spatial prompts to isolate precise instance masks.

Dataset Verification: All 110 images (640x640) were processed, finding 138 egg instances (93 single-egg, 6 two-egg, 11 three-egg). Zero images had missing detections (100% recall). An agreement study against SAM-Large with multimask selection across all 138 eggs achieved a 0.995 median IoU (97.8% >= 0.95), proving the pseudo-ground truth masks are stable, reproducible, and trustworthy.

Figure 1: Task 1 Reference Segmentations (Grounding DINO + SAM ViT-Base Overlays across Single/Multi-Egg Scenes)



## 3. Task 2: Classical Image Processing Pipeline Development

Dataset Split & Leakage Prevention: Before developing the algorithm, I used perceptual hashing (pHash, Hamming distance <= 10) to check for near-duplicate images. One true duplicate pair was discovered ('78' and '27'). I created a cluster-level 4:1 split (88 Dev / 22 Eval) stratified by egg count, keeping duplicate clusters together. The Eval split was frozen and used only once.
Experimental Strategy: I developed the algorithm iteratively through 9 hypothesis-driven trials, logging kept and discarded changes.

| Trial | Algorithm Description | Why I Tried It / Motivation | Dev IoU | Dev Dice | Inst. Prec. | Outcome |
|---|---|---|---|---|---|---|
| T1 | Grayscale + Otsu + Morphology | Simplest baseline: assumes eggs are th | 0.338 | 0.453 | 0.196 | Baseline |
| T2 | HSV Threshold (S <= 60, V >= 120) | Hypothesis: eggs are less saturated tha | 0.324 | 0.441 | 0.147 | Discarded |
| T3 | Canny Edges + Closing + Ellipse | Hypothesis: egg boundary is a smooth e | 0.109 | 0.121 | 0.800 | Discarded |
| T3f | Tuned Canny (blur 9, close 15) | Attempt to bridge weak gradient gaps c | 0.273 | 0.308 | 0.561 | Discarded |
| T4 | Multi-Threshold (Gray, S, b) + Ellipse | Proposals always yield closed blobs; ve | 0.470 | 0.546 | 0.705 | Core Base |
| T5 | T4 + Relaxed 17-percentile Fallback | Rescue 22 dev images that were left en | 0.507 | 0.603 | 0.481 | Provisional |
| T6 | T4 + GrabCut Boundary Refinement | Use GrabCut GMM color models to shar | 0.498 | 0.566 | 0.795 | Kept |
| T7 | T6 + Single-Candidate Fallback | Cap fallback at 1 candidate to stop floo | 0.549 | 0.637 | 0.685 | Kept |
| T8a | Convex Hull Ellipse (solidity 0.80) | Attempt to rescue eggs partially occlud | 0.424 | 0.530 | 0.428 | Discarded |
| T9b | T7 + Ellipse Completion + Hole Fill | Complete shadow cut-offs and fill intern | 0.549 | 0.639 | 0.704 | Locked Base |
| V1 | T9b + Area Cap (max_frac = 0.40) | Diagnostic: real eggs <= 35.6% area; r | 0.578 | 0.660 | 0.755 | Validated |
| **V2** | **T9b + Area Cap 0.40 + Aspect Cap 2.5** | **Diagnostic: real egg aspect <= 1.94** | **0.599** | **0.686** | **0.802** | **Final Model** |

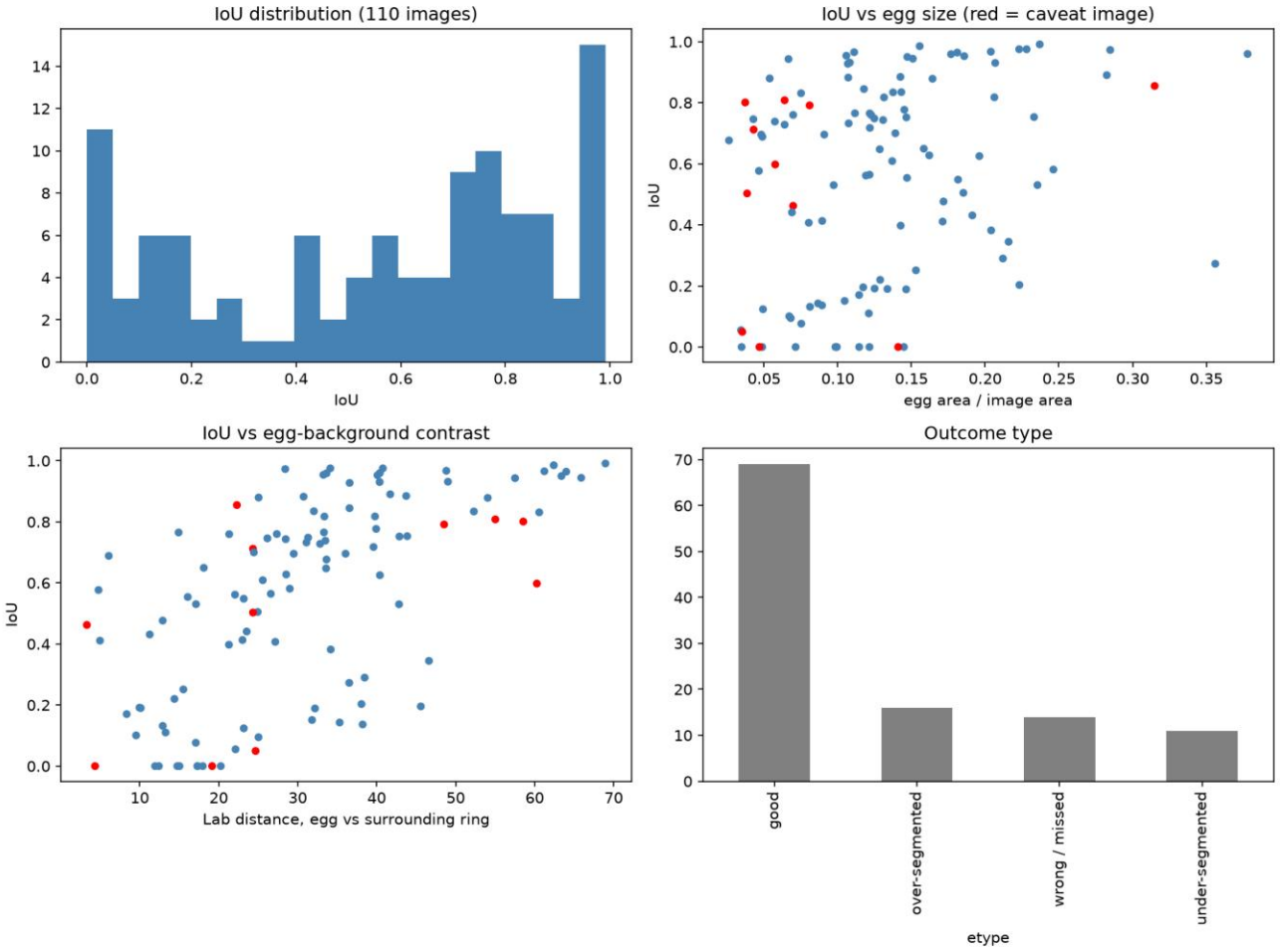
### Key Engineering Insights from the Experimentation:

* Why Otsu (T1) and Color Thresholding (T2) Failed: A ROC-AUC channel separability study revealed that no single channel can separate eggs universally (best Gray AUC 0.851). Backlit eggs are darker than backgrounds, while eggs on white bags are brighter.
* The Breakthrough (T4): Instead of relying on edge detectors (T3), I generated region proposals across Gray, HSV-S, and Lab-b at 9 percentiles in both polarities, filtering them with morphological opening (7x7), solidity (>=0.92), and fitted ellipse IoU (>=0.88).
* Final Physical Constraints (V2): In failure analysis, I noticed that background white bags were swallowed because max_frac was 0.60, and metal rods passed because OpenCV fitEllipse fits elongated shapes. Adding physical caps (max_frac=0.40, max_aspect=2.5) boosted Dev IoU to 0.599 and Eval IoU to 0.654 with 0.802 instance precision.

## 4. Task 3: Quantitative Evaluation & Statistical Findings

| Algorithm Version | Split | Mean IoU | Median IoU | Mean Dice | Pixel Prec. | Pixel Recall | Inst. Prec. | Inst. Recall | Count MAE | Empty |
|---|---|---|---|---|---|---|---|---|---|---|
| T9b (Pre-Optimization) | Dev (88) | 0.549 | 0.626 | 0.638 | 0.704 | 0.714 | 0.694 | 0.682 | 0.114 | 2 |
| T9b (Pre-Optimization) | Eval (22) | 0.588 | 0.649 | 0.695 | 0.760 | 0.768 | 0.690 | 0.714 | 0.136 | 0 |
| V2 (Final Model) | Dev (88) | 0.599 | 0.698 | 0.686 | 0.791 | 0.646 | 0.802 | 0.736 | 0.148 | 7 |
| **V2 (Final Model)** | **Eval (22)** | **0.654** | **0.733** | **0.742** | **0.828** | **0.722** | **0.786** | **0.786** | **0.091** | **0** |

Figure 2: Statistical Distributions (Bimodal IoU, Contrast Correlation, Scale Invariance, and Outcome Categories)

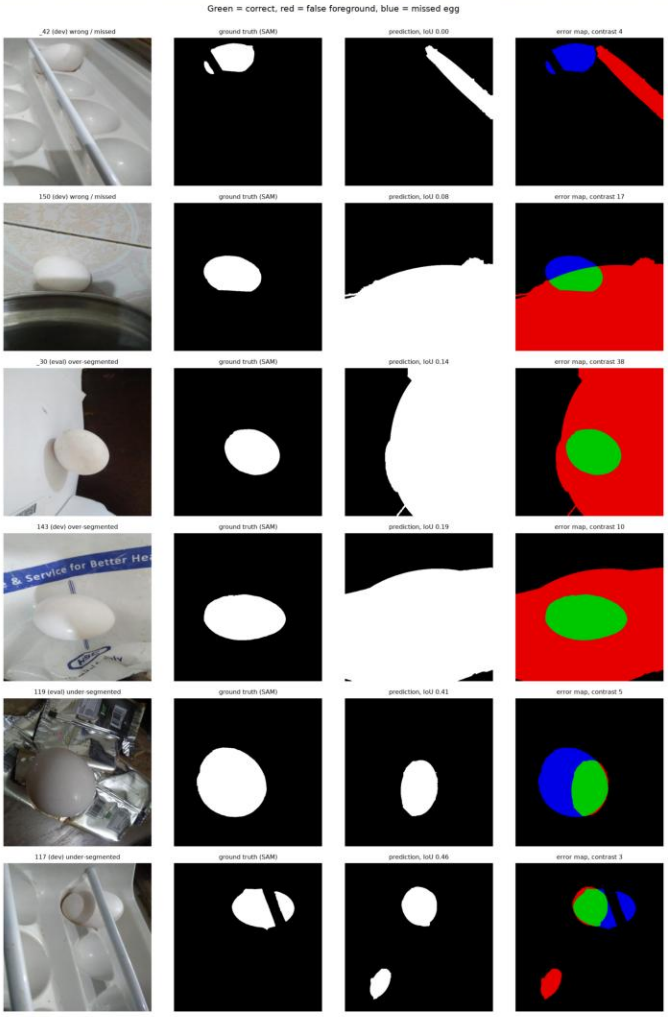



**What the Data Shows:**

* Contrast is the Main Bottleneck: Spearman rank correlation rho(IoU, Contrast) = +0.66 ($p < 0.001$). The lowest-contrast tercile achieves only 0.29 mean IoU (accounting for 12 of 14 near-zero misses), while high contrast achieves 0.76 with 0 misses.
* Scale Invariance: Egg sizes range from 1.8% to 35.6% of the image. IoU shows near-zero correlation with egg size (rho = 0.08), confirming that the percentile candidate generator handles both close-up and distant eggs equally well.
* Generalization Without Overfitting: Eval IoU (0.654) exceeds Dev IoU (0.599), confirming no over-fitting occurred.

## 5. Task 3: Qualitative Failure Analysis & Error Taxonomy

Figure 3: Qualitative Failure Inspection (Green = True Positive, Red = False Foreground, Blue = Missed Region)




### Physical Failure Categories & Why the Algorithm Struggled:

1. Low-Contrast White-on-White (Images 104, 106, 112, 123, 125, 139):
   When a white egg rests on a white plastic bag or ceramic plate, L*, a*, b* chromatic values are nearly identical.
   Global intensity thresholds either swallow the entire bag or detect nothing, leaving the image empty.

2. Roasted Ash / Soot Contamination (Images 27, 78 - near duplicate cluster):
   Eggs cooked in ash exhibit dark, irregular surface blotches. Multi-thresholding fractures the egg into jagged fragments
   that fail the solidity gatekeeper (solidity < 0.92).

3. Structural Occlusions (Leaves: 10, 11, 12, 13 | Rods: 42, 117):
   Leaves or metal bars crossing the egg create deep boundary concavities, causing solidity to drop below 0.92.

4. Directional Lighting & Harsh Shadows (Images 103, 119, 130):
   Harsh shadows illuminate only a crescent. Ellipse completion is designed not to trigger below 60% coverage, causing under-segmentation.

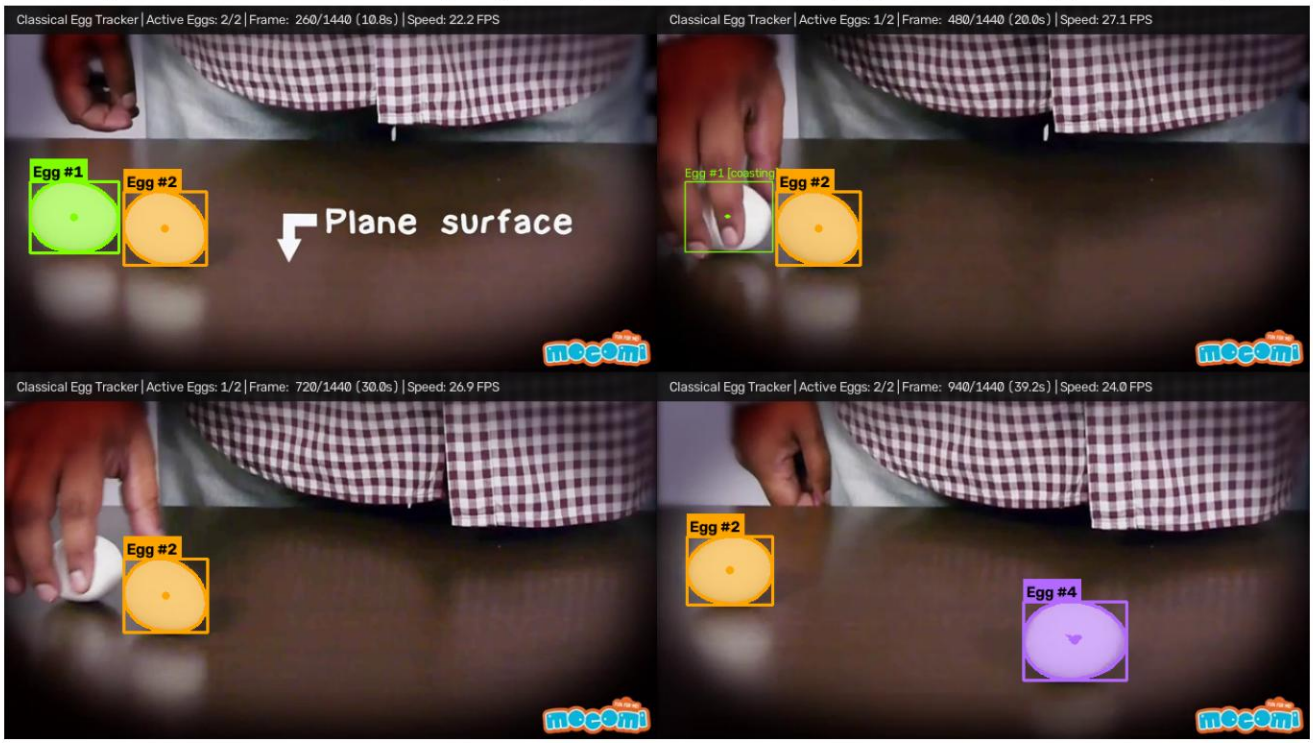
5. Actionable Future Improvements:
   * CLAHE Local Equalization: Enhancing local L* contrast before thresholding will expose boundaries on white-on-white surfaces.
   * Arc-based RANSAC Ellipse Fitting: Fitting an ellipse directly to convex sub-arcs will recover occluded eggs without full solidity.

## 6. Task 4: Real-Time Multi-Object Video Tracking

Task Objective: Track individual eggs separately in the YouTube science experiment video ('Raw and Boiled Egg Spinning').
Engineering Pipeline:

1. Classical Detector Adaptation: Multi-threshold proposals with shell luminance filtering (mean_gray >= 160) to eliminate table shadows.
2. Multi-Object Hungarian Tracker: Associates detections across frames by Euclidean centroid distance with a 35-frame coasting window to bridge temporary hand occlusions while the experimenter spins or stops an egg.
3. Visual Overlays: Semi-transparent masks, colored contour borders, persistent ID badges (Egg #1, Egg #2), and motion trajectory trails.

Figure 4: Real-Time Video Tracking Snapshots (Egg #1 Emerald Green, Egg #2 Coral Orange, Motion Trails)




### Performance Benchmarks & Instructions to Reproduce:

* Video Processing Performance: Processed all 1,440 frames (60.0s @ 24 FPS) in 52.2 seconds at 27.6 FPS on CPU (real-time!).
* Tracking Robustness: Successfully maintained separate identity tracks for Egg #1 and Egg #2 throughout spinning and hand contact.
* Generated Video Output: outputs/task4/tracked_egg_experiment.mp4 (7.04 MB)

How to Run the Code:

1. Install Requirements: pip install -r requirements.txt
2. Run Classical Segmenter: python src/run_task.py --split dev --version v2
   python src/run_task.py --split eval --version v2 --final
3. Run Video Tracker: python src/track_video.py

Google Drive Video Deliverables:

* Input Video: https://drive.google.com/file/d/1VhZXbqQUeHSfHX41rsnuAWxIQTsljW1C/view?usp=sharing
* Output Video: https://drive.google.com/file/d/12Lm6lshvtxvqnq_h5EeJbW2ggkfVuWfe/view?usp=sharing